



Data
Divas



Intelligent Candidate Ranking System

INDIA RUNS - Data & AI
Challenge | Track 1: Data & AI

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The Problem



Recruiters go through hundreds of profiles
and still miss the right person.

Not because the talent isn't there —
but because keyword filters can't see
what actually matters.

We built a smarter way to rank candidates
the way a great recruiter thinks.

Our Solution



A Python-based intelligent ranking system
that scores 100,000 candidates in minutes.

No LLMs. No black box.

Just smart, explainable scoring logic
that a recruiter can trust.

Output: Top 100 best-fit candidates
ranked with clear reasoning.

Scoring Logic



- ▶ Title Relevance — 30% ML Engineer, AI Engineer, Data Scientist = high score HR Manager, Accountant = disqualified
- ▶ Career History — 20% Rewards product company experience Penalizes pure consulting backgrounds
- ▶ Years of Experience — 15% Sweet spot: 5-9 years gets full marks
- ▶ Skills Match — 20% Core AI/ML skills: Python, NLP, PyTorch, Embeddings, Vector Search, LLMs Trust weighted by duration + endorsements
- ▶ Behavioral Signals — 10% Last active date, open to work, recruiter response rate, notice period
- ▶ Location — 5%
Prefers Pune, Noida, Bangalore, Hyderabad

Results

100,000 candidates processed
Top 100 ranked and shortlisted
Validation passed with 0 errors

Top candidates identified:

- Senior Machine Learning Engineer — 7.2 yrs —
Score: 0.9756
- Senior Data Scientist — 7.4 yrs — Score: 0.9648
- Machine Learning Engineer — 5.7 yrs — Score:
0.9630
- Staff Machine Learning Engineer — 7.0 yrs —
Score: 0.9587
- Lead AI Engineer — 6.7 yrs — Score: 0.9575

Score range: 0.9756 → 0.8685

Runtime: Under 2 minutes on CPU

Tech Stack



Language:
Python 3.12

Libraries:

pandas — data processing
numpy — numerical operations
json — reading candidate profiles
datetime — behavioral signal scoring



Dataset:

100,000 candidate profiles
in JSONL format

Tools:

GitHub — version control & code hosting
Canva — presentation
Windows — local execution environment

Runtime:

~2 minutes for 100,000 candidates
on standard laptop CPU

Challenges We Faced



Language:

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How We Handled Traps



- ◆ Keyword Stuffers
If 15+ skills all marked
"expert"
→ score penalized by 40%

- ◆ Irrelevant Job Titles
HR Manager, Accountant,
Teacher
→ automatically
disqualified
score capped at 0.15

- ◆ Inactive Candidates
Last active over 180 days ago
→ zero recency score

- ◆ Pure Consulting Backgrounds
TCS, Infosys, Wipro only experience
→ career score reduced by 60%

- ◆ Tie Breaking
Equal scores sorted by
candidate ID ascending
as per validator rules

- ◆ Semantic Matching

Use NLP to understand job descriptions
beyond simple keyword matching

- ◆ LLM Integration

Use language models to read and
understand candidate summaries deeply

- ◆ Dynamic Weightage

Allow recruiters to adjust scoring
weights based on their priorities

- ◆ Real Time Ranking

Build a web interface where recruiters
can upload JD and get instant rankings

- ◆ Multi Role Support

Currently built for Senior AI Engineer
— expand to any job role automatically

Future Improvements





Intelligent Candidate Ranking System

Built for INDIA RUNS — Data & AI Challenge
by DataDivas

Thank You



GitHub:

github.com/sowmyajadala/India-runs-ranker

"Hiring is broken. We're here to fix it —
one ranked candidate at a time."

— Sowmya Jadala

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